Exp No:18 Experiment to understand KNN algorithm for a given data

Set

AIM:

To implement the k-nearest neighbours classification algorithm to satisfy fruits based on their weight and sweetness.

Description:

Understand the KNN algorithm for the dataset given.

CODE:

```
import numpy as np

import pandas as pd

df=pd.read_csv('Iris.csv')

df.info()

df.variety.value_counts()

df.head()

features=df.iloc[:,:-1].values

label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,rando

model_KNN=KNeighborsClassifier(n_neighbors=5)

model_KNN.fit(xtrain,ytrain)

print(model_KNN.score(xtrain,ytrain))

print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix

confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report

print(classification_report(label,model_KNN.predict(features)))
```

OUTPUT:

1. Dataset Information

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 3 columns):

#	Column	Non-Null Count	Dtype
0	Weight	30 non-null	int64
1	Sweetness	30 non-null	float64
2	Fruit	30 non-null	object

dtypes: float64(1), int64(1), object(1)
memory usage: 852.0+ bytes

2. Fruit Class Distribution

Fruit
Apple 10
Banana 10
Orange 10
Name: count, dtype: int64

3. Original Dataset – First Five Records

	Weight	Sweetness	Fruit
0	120	6.0	Apple
1	125	6.2	Apple
2	130	6.4	Apple
3	135	6.5	Apple
4	140	6.7	Apple

4. Label Encoding

	Weight	Sweetness	Fruit
0	120	6.0	0
1	125	6.2	0
2	130	6.4	0
3	135	6.5	0
4	140	6.7	0

5. KNN Model

KNeighborsClassifier()

6. Training and Testing Accuracy

Training Accuracy: 0.9166666666666666
Testing Accuracy: 0.6666666666666666

7. Confusion Matrix

```
array([[ 9,  0,  1],
       [ 3,  7,  0],
       [ 0,  0, 10]])
```

8. Classification Report

	precision	recall	f1-score	support
0	0.75	0.90	0.82	10
1	1.00	0.70	0.82	10
2	0.91	1.00	0.95	10
accuracy			0.87	30
macro avg	0.89	0.87	0.86	30
weighted avg	0.89	0.87	0.86	30

RESULT:

The KNN classification algorithm was successfully implemented and the fruits were classified based on their features. The model performance was evaluated using accuracy confusion matrix and classification repeat.